RevoGrocers Sales Performance Analysis

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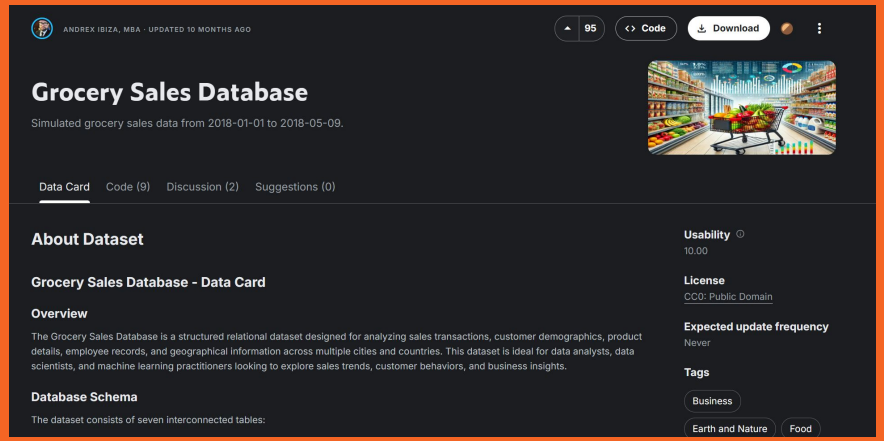
This analysis is based on a public dataset from Kaggle:



[LINK DATASET]

RevoGrocers is a fictional company created for analytical purposes

Any insights or recommendations are based on this dataset and do not reflect real-world data

BUSINESS CONTEXT

RevoGrocers is a fictional grocery retail business that operates in multiple locations, offering a diverse range of grocery products to customers. The company aims to optimize sales strategies, enhance customer experience, and increase revenue by leveraging data-driven decision-making.

Identify the product category that generates the highest revenue after discount

```
SELECT
    c.categoryname,
    SUM((p.Price * s.Quantity) - COALESCE(s.Discount,
0)) AS TotalRevenue
FROM `fsda-sql-01.grocery_dataset.sales` s
  JOIN `fsda-sql-01.grocery_dataset.products` p
  ON s.ProductID = p.ProductID
    JOIN `fsda-sql-01.grocery_dataset.categories` c
    ON p.categoryid = c.categoryid
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

categoryname	TotalRevenue
Confections	$574,077,762.31
Meat	$508,091,936.15
Poultry	$453,592,801.05
Cereals	$440,629,310.47
Snails	$383,585,174.24
Beverages	$377,860,409.92
Produce	$374,112,765.57
Dairy	$365,209,022.81
Seafood	$340,707,813.90
Grain	$333,850,879.60
Shell fish	$308,794,104.97

The **Confections** category is the clear top performer, generating **$574,077,762.31** in revenue.

Assess the relation between revenue after discount and total units sold for each product category

```sql
SELECT
    c.categoryname,
    SUM(s.Quantity) AS TotalUnitsSold,
    SUM((p.Price * s.Quantity) - COALESCE(s.Discount, 0))
AS TotalRevenue,
    ROUND(SUM((p.Price * s.Quantity) - COALESCE(s.Discount,
0)) / NULLIF(SUM(s.Quantity), 0),2) AS AvgRevenuePerUnit
FROM `fsda-sql-01.grocery_dataset.sales` s
  JOIN `fsda-sql-01.grocery_dataset.products` p
  ON s.ProductID = p.ProductID
    JOIN `fsda-sql-01.grocery_dataset.categories` c
    ON p.categoryid = c.categoryid
GROUP BY 1
ORDER BY 2 DESC;
```

categoryname	TotalUnitsSold	TotalRevenue	AvgRevenuePerUnit
Confections	11078474	$574,077,762.31	$51.82
Meat	9719292	$508,091,936.15	$52.28
Poultry	9159847	$453,592,801.05	$49.52
Cereals	8735296	$440,629,310.47	$50.44
Produce	8174673	$374,112,765.57	$45.76
Beverages	7393693	$377,860,409.92	$51.11
Snails	7199409	$383,585,174.24	$53.28
Seafood	6996152	$340,707,813.90	$48.7
Shell fish	6983457	$308,794,104.97	$44.22
Dairy	6815143	$365,209,022.81	$53.59
Grain	5433152	$333,850,879.60	$61.45

Confections is the "**Cash Cow**" with high volume and a healthy price point.

Find the relation between revenue after discount and the number of unique customers for each product category

```sql
SELECT
    c.categoryname,
    COUNT(DISTINCT s.CustomerID) AS UniqueCustomers,
    ROUND(SUM((p.Price * s.Quantity) - COALESCE(s.Discount,
0)) / COUNT(DISTINCT s.CustomerID), 2) AS
AvgSpendPerCustomer
FROM `fsda-sql-01.grocery_dataset.sales` s
  JOIN `fsda-sql-01.grocery_dataset.products` p
  ON s.ProductID = p.ProductID
    JOIN `fsda-sql-01.grocery_dataset.categories` c
    ON p.categoryid = c.categoryid
GROUP BY 1
ORDER BY 2 DESC;
```

categoryname	UniqueCustomers	AvgSpendPerCustomer
Confections	98743	$5,813.86
Meat	98701	$5,147.79
Poultry	98679	$4,596.65
Cereals	98651	$4,466.55
Produce	98601	$3,794.21
Beverages	98424	$3,839.11
Snails	98376	$3,899.17
Shellfish	98338	$3,140.13
Seafood	98334	$3,464.80
Dairy	98308	$3,714.95
Grain	97335	$3,429.92

The **Confections** category is the clear top performer, Averaging $5,813.86 across 98743 Unique Customers

Calculate the average price unit for each product category in catalogue

```sql
SELECT
    c.CategoryName,
    COUNT(p.productid) AS TotalPoductsInCatalogue,
    ROUND(AVG(p.price), 2) AS AvgCataloguePrice
FROM `fsda-sql-01.grocery_dataset.products` p
    JOIN `fsda-sql-01.grocery_dataset.categories` c
    ON p.categoryid = c.categoryid
WHERE p.price IS NOT NULL
GROUP BY 1
ORDER BY 3 DESC;
```

CategoryName	TotalPoductsInCatalogue	AvgCataloguePrice
Grain	28	$61.40
Dairy	35	$53.56
Snails	37	$53.20
Meat	50	$52.27
Confections	57	$51.85
Beverages	38	$51.04
Cereals	45	$50.42
Poultry	47	$49.46
Seafood	36	$48.67
Produce	42	$45.80
Shellfish	36	$44.27

Confections is the "Anchor," proving customers accept the **~$51** price point with 57 Variety of Products.

Evaluate the relation between the average price per unit and the number of buyers (unique customers) per category

```sql
SELECT
    c.CategoryName,
    ROUND(AVG(p.Price), 2) AS AvgPricePerUnit,
    COUNT(DISTINCT s.CustomerID) AS UniqueBuyers,
    ROUND(SUM((p.Price * s.Quantity) - COALESCE(s.Discount,
0)) / COUNT(DISTINCT s.CustomerID), 2) AS
AvgRevenuePerBuyer
FROM `fsda-sql-01.grocery_dataset.sales` s
  JOIN `fsda-sql-01.grocery_dataset.products` p
  ON s.ProductID = p.ProductID
    JOIN `fsda-sql-01.grocery_dataset.categories` c
    ON p.categoryid = c.categoryid
WHERE p.price IS NOT NULL
GROUP BY 1
ORDER BY 2 DESC;
```

CategoryName	AvgPricePerUnit	UniqueBuyers	AvgRevenuePerBuyer
Grain	$61.43	97335	$3,429.92
Dairy	$53.61	98308	$3,714.95
Snails	$53.28	98376	$3,899.17
Meat	$52.31	98701	$5,147.79
Confections	$51.81	98743	$5,813.86
Beverages	$51.13	98424	$3,839.11
Cereals	$50.44	98651	$4,466.55
Poultry	$49.51	98679	$4,596.65
Seafood	$48.68	98334	$3,464.80
Produce	$45.79	98601	$3,794.21
Shell fish	$44.23	98338	$3,140.13

Confections sits in the middle price range ($51.81) but achieves the **Highest Buyer Count (98,743)** and substantially highest **Spend Per Buyer ($5,813)**.

```sql
WITH CategoryStats AS (
    SELECT
        c.CategoryName,
        SUM(
          CASE
            WHEN s.TotalPrice = 0 OR s.TotalPrice IS NULL
            THEN (p.price * s.quantity) - COALESCE(s.Discount, 0)
            ELSE s.TotalPrice
          END
        ) AS Revenue
    FROM `fsda-sql-01.grocery_dataset.sales` s
      JOIN `fsda-sql-01.grocery_dataset.products` p
      ON s.ProductID = p.ProductID
        JOIN `fsda-sql-01.grocery_dataset.categories` c
        ON p.categoryid = c.categoryid
    GROUP BY 1
),
TotalStoreStats AS (
  SELECT SUM(Revenue) AS GrandTotal FROM CategoryStats
)
SELECT
    cs.CategoryName,
    cs.Revenue AS CategoryRevenue,
    ROUND((cs.Revenue / ts.GrandTotal)* 100, 2) AS RevenuePercentage
FROM CategoryStats cs
CROSS JOIN TotalStoreStats ts
ORDER BY RevenuePercentage DESC;
```

Which categories contribute the most to overall revenue after discount (percentage-wise)?

CategoryName	CategoryRevenue	RevenuePercentage
Confections	$574,077,762.31	12.87%
Meat	$508,091,936.15	11.39%
Poultry	$453,592,801.05	10.17%
Cereals	$440,629,310.47	9.88%
Snails	$383,585,174.24	8.60%
Beverages	$377,860,409.92	8.47%
Produce	$374,112,765.57	8.39%
Dairy	$365,209,022.81	8.19%
Seafood	$340,707,813.90	7.64%
Grain	$333,850,879.60	7.48%
Shell fish	$308,794,104.97	6.92%

Confections is the primary revenue driver, contributing **12.87%** of total sales ($574M).

```sql
WITH CustomerCategoryHistory AS (
  SELECT
      c.categoryname,
      s.CustomerID,
      COUNT(DISTINCT s.TransactionNumber) AS NumberOfVisits
  FROM `fsda-sql-01.grocery_dataset.sales` s
        JOIN `fsda-sql-01.grocery_dataset.products` p
        ON s.ProductID = p.ProductID
          JOIN `fsda-sql-01.grocery_dataset.categories` c
          ON p.categoryid = c.categoryid
  GROUP BY 1,2)
SELECT
    CategoryName,
    COUNT(CustomerID) AS TotalCustomers,
    SUM(CASE WHEN NumberOfVisits > 1 THEN 1 ELSE 0 END) AS
ReturningCustomers,
    ROUND((SUM(CASE WHEN NumberOfVisits > 1 THEN 1 ELSE 0 END) *
100.0) / COUNT(CustomerID),2) AS RepeatPurchaseRate_Percent
FROM CustomerCategoryHistory
GROUP BY CategoryName
ORDER BY 4 DESC;
```

Which product categories have the highest repeat purchase rate?

CategoryName	TotalCustomers	ReturningCustomers	RepeatPurchaseRate_Percent
Confections	98743	98598	99.85%
Meat	98701	98318	99.61%
Poultry	98679	98122	99.44%
Cereals	98651	97867	99.21%
Produce	98601	97550	98.93%
Beverages	98424	96679	98.23%
Snails	98376	96324	97.91%
Seafood	98334	96138	97.77%
Shellfish	98338	96054	97.68%
Dairy	98308	95677	97.32%
Grain	97335	91184	93.68%

Confections leads with **99.85%**, meaning virtually every single customer who bought candy once came back to buy it again.

The most critical finding is that **RevoGrocers is a B2B / Wholesale Distributor**, not a standard B2C supermarket.

- **Evidence:**
  - **High Price Floor:** The cheapest category averages **~$44/unit**.
  - **High Spend:** Customers spend **$3,000 - $5,800** per category.
  - **Market Penetration:** Every category has nearly identical reach (~98,000 unique buyers).
  - **Retention:** Repeat purchase rates are near-perfect (>93%), indicating contractual supply relationships rather than casual shopping.

Confections

Performance: It is the undisputed leader. It ranks **#1 in Revenue ($574M)**, **#1 in Volume (11M units)**, and **#1 in Loyalty (99.85%)**.
Insight: Despite being a "want" rather than a "need," it is the anchor of your business. Clients are allocating the highest share of their wallet ($5,813/buyer) to this category.

Meat, Poultry, Cereals

Performance: These categories are consistent performers, holding the #2, #3, and #4 spots respectively.
Insight: Together with Confections, the top 3 categories generate **~34.4%** of total revenue. This is your stable "Protein & Sugar" core.

Grain & Shellfish

Grain (High Price Friction): It has the highest price (**$61.43**) but the lowest transaction count. The high cost is acting as a barrier, causing the lowest (though still high) retention rate of 93%.
Shell fish (Low Value): It has the lowest price (**$44.23**) but fails to drive volume. Being "cheap" is not helping it sell. It generates the lowest revenue per customer ($3,140).

Find the cumulative amount of transaction of the top user (user with highest transaction value, window function)

```
WITH TopUser AS (
    SELECT
        s.CustomerID
    FROM `fsda-sql-01.grocery_dataset.sales` s
       JOIN `fsda-sql-01.grocery_dataset.products` p
       ON s.ProductID = p.ProductID
            JOIN `fsda-sql-01.grocery_dataset.categories` c
            ON p.categoryid = c.categoryid
    GROUP BY 1
    ORDER BY SUM(
          CASE
            WHEN s.TotalPrice = 0 OR s.TotalPrice IS NULL
            THEN (p.price * s.Quantity)
            ELSE s.TotalPrice
          END) DESC
    LIMIT 1),
```

```sql
UserHistory AS (
  SELECT
      s.CustomerID,
      c.categoryname,
      s.TransactionNumber,
      s.SalesDate,
      SUM(
        CASE WHEN s.TotalPrice = 0 OR s.TotalPrice IS NULL
          THEN (p.price * s.Quantity)
          ELSE s.TotalPrice
        END) AS TransactionValue
  FROM `fsda-sql-01.grocery_dataset.sales` s
    JOIN `fsda-sql-01.grocery_dataset.products` p
    ON s.ProductID = p.ProductID
      JOIN `fsda-sql-01.grocery_dataset.categories` c
      ON p.categoryid = c.categoryid
  WHERE s.CustomerID = (SELECT CustomerID From TopUser)
  GROUP BY 1,2,3,4)
```

```sql
SELECT
    CustomerID,
    SalesDate,
    CategoryName,
    TransactionNumber,
    TransactionValue AS DailySpend,
    SUM (TransactionValue) OVER (
      ORDER BY SalesDate
      ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ) AS CumulativeSpend
FROM UserHistory
ORDER BY SalesDate;
```

CustomerID	CategoryName	DailySpend	CumulativeSpend
94800	Seafood	2013.0048	2013.0048
94800	Beverages	2156.5752	4169.58
94800	Dairy	2366.3472	6535.9272
94800	Seafood	2251.3128	8787.24
94800	Snails	342.4512	9129.6912
94800	Seafood	377.3592	9507.0504
94800	Meat	1166.3544	10673.4048
94800	Meat	2245.4712	12918.876
94800	Grain	1501.104	14419.98
94800	Cereals	1867.3992	16287.3792
94800	Cereals	1111.8048	17399.184
94800	Cereals	2313.9048	19713.0888
94800	Shell fish	1524.9336	21238.0224
94800	Meat	1038.8736	22276.896
94800	Dairy	525.1344	22802.0304
94800	Poultry	1113.6792	23915.7096
94800	Snails	1054.0368	24969.7464
94800	Snails	585.3912	25555.1376
94800	Seafood	1629.576	27184.7136
94800	Meat	482.4216	27667.1352
94800	Grain	2313.2904	29980.4256
94800	Poultry	764.088	30744.5136
94800	Meat	2162.7768	32907.2904
94800	Beverages	2357.9304	35265.2208
94800	Produce	640.944	35906.1648
94800	Seafood	411.0576	36317.2224
94800	Snails	2109.6552	38426.8776
94800	Seafood	1.0776	38427.9552
94800	Seafood	2296.8576	40724.8128
94800	Meat	1829.76	42554.5728
94800	Grain	1280.7216	43835.2944
94800	Dairy	1242.6096	45077.904
94800	Confections	1334.5416	46412.4456
94800	Dairy	1307.1432	47719.5888
94800	Beverages	1845.5712	49565.16
94800	Shell fish	163.9968	49729.1568
94800	Confections	1866.7296	51595.8864
94800	Poultry	1677.9912	53273.8776
94800	Snails	1328.8392	54602.7168
94800	Snails	1093.3224	55696.0392
94800	Snails	2109.6552	57805.6944
94800	Seafood	411.0576	58216.752
94800	Snails	1963.0632	60179.8152
94800	Shell fish	57.336	60237.1512
94800	Produce	146.3424	60383.4936
94800	Dairy	1307.1432	61690.6368
94800	Confections	499.6056	62190.2424
94800	Produce	1892.676	64082.9184
94800	Confections	1670.1144	65753.0328
94800	Beverages	182.4456	65935.4784
94800	Produce	2004.9336	67940.412
94800	Confections	1866.7296	69807.1416
94800	Shell fish	1340.4744	71147.616
94800	Grain	1421.9208	72569.5368
94800	Seafood	669.492	73239.0288
94800	Meat	806.1192	74045.148
94800	Produce	1016.7168	75061.8648
94800	Shell fish	1078.596	76140.4608
94800	Cereals	2397.012	78537.4728
94800	Meat	254.3136	78791.7864
94800	Confections	212.2296	79004.016
94800	Meat	1707.7224	80711.7384
94800	Snails	2210.0928	82921.8312
94800	Dairy	1362.9552	84284.7864
94800	Seafood	1162.068	85446.8544
94800	Confections	2244.9216	87691.776
94800	Snails	1093.3224	88785.0984
94800	Cereals	814.776	89599.8744
94800	Meat	254.3136	89854.188
94800	Snails	1646.3904	91500.5784
94800	Poultry	1316.9256	92817.504
94800	Confections	1221.2088	94038.7128
94800	Snails	1937.1192	95975.832
94800	Grain	1501.104	97476.936
94800	Meat	1038.8736	98515.8096
94800	Produce	70.3848	98586.1944
94800	Confections	2244.9216	100831.116
94800	Produce	1647.312	102478.428
94800	Cereals	814.776	103293.204
94800	Cereals	31.0944	103324.2984
94800	Seafood	594.7416	103919.04
94800	Produce	70.3848	103989.4248
94800	Beverages	2357.9304	106347.3552
94800	Grain	2316.48	108663.8352
94800	Confections	1745.8968	110409.732
94800	Cereals	1867.3992	112277.1312
94800	Meat	518.7192	112795.8504
94800	Shell fish	1481.904	114277.7544
94800	Snails	2017.3584	116295.1128
94800	Seafood	885.996	117181.1088
94800	Produce	711.2928	117892.4016
94800	Confections	778.608	118671.0096
94800	Confections	1653.0936	120324.1032
94800	Dairy	1382.4312	121706.5344
94800	Produce	1452.7536	123159.288
94800	Confections	1562.1792	124721.4672
94800	Seafood	944.3904	125665.8576
94800	Confections	904.2984	126570.156
94800	Shell fish	2255.4048	128825.5608
94800	Poultry	1416.0792	130241.64
94800	Seafood	2325.0744	132566.7144
94800	Shell fish	57.336	132624.0504

94800	Seafood	1785.7824	134409.8328

Customer **#94800** is your top buyer, with a visible cumulative spend of **$134,409.83**.

Actionable Recommendations

1. **Optimize "Grain" Packaging:** The **$61 price point** is the only visible friction point in your data. Introduce a smaller SKU (e.g., a "Half-Sack" at $30) to capture smaller businesses that cannot afford the bulk option.
2. **Bundle "Shell fish":** Since Shell fish has low value but high reach, stop selling it as a standalone item. Create **"Surf & Turf" bundles** (Meat + Shell fish) to force higher volume sales of the under-performing seafood.
3. **Subscription Model:** With Repeat Purchase Rates over **93%**, you should transition customers to **Automatic Reordering**. This locks in revenue and prevents competitors from stealing your highly loyal base.